Civics Group | Index Number

Candidate's Name: ______________________________________

MERIDIAN JUNIOR COLLEGE
Preliminary Examinations 2011
Higher 2

H2 BIOLOGY **9648/03**

Applications Paper and Planning Question **21 September 2011**

Paper 3 **2 hours**

Additional Materials: Answer papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough workings.
Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Answer **all** questions.
At the end of the examination,
1. Fasten all your writing papers for Question 4 and Question 5 separately.
2. Hand in the following separately:
    - Question 1 – 3
    - Question 4
    - Question 5

The number of marks is given in brackets [ ] at the end of each question or part question.

For examiner's Use	
1	/ 15
2	/ 14
3	/ 11
4	/ 12
5	/ 20
Total	**/ 72**

This paper consists of **10** printed pages.

[Turn over]

For Examiner's Use

Answer **all** questions.

1. The synthesis of caffeine in coffee plants involves enzymes which add methyl groups (CH_3) to convert xanthosine to caffeine:

$$\textbf{xanthosine} \xrightarrow[+\,CH_3]{\text{theobromine synthase}} \textbf{theobromine} \xrightarrow[+\,CH_3]{\text{caffeine synthase}} \textbf{caffeine}$$

In an attempt to produce caffeine-free coffee, cells of a coffee plant, *Coffea canephora*, were grown in tissue culture and genetically modified to suppress expression of the gene for theobromine synthase.

DNA was constructed to code either for short or for long lengths of RNA with base sequences complementary to parts of the mRNA produced by the gene for theobromine synthase.

(a) Explain how lengths of RNA that are complementary to mRNA may suppress the expression of the gene for theobromine synthase. [3]

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Three types of cell were then cloned in tissue culture into plantlets:

- **A** - unmodified cells
- **B** - genetically modified cells with DNA that codes for short lengths of RNA complementary to mRNA for theobromine synthase
- **C** - genetically modified cells with DNA that codes for long lengths of RNA complementary to mRNA for theobromine synthase.

Samples of each of the three types of plantlet were analysed to measure their theobromine and caffeine content. The results of the analysis are shown in Fig. 1.1.

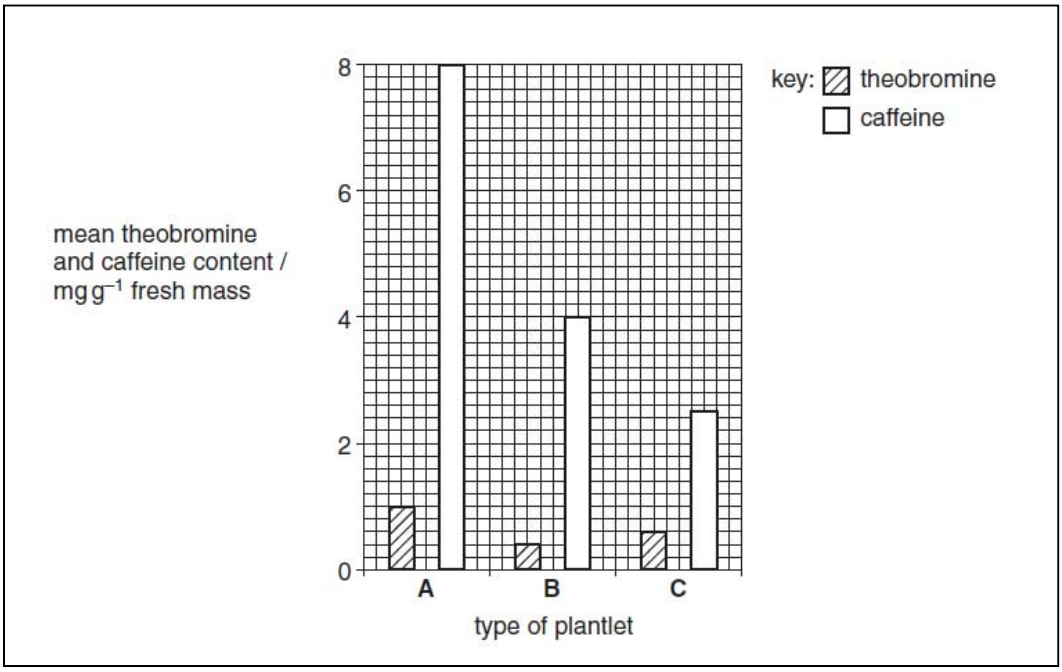

Fig. 1.1

(b) With reference to Fig. 1.1, describe and explain the results of the experiment. [5]

For Examiner's Use

(c) Describe briefly how plants are cloned by tissue culture. [4]

(d) Explain the advantages of using cloned plants in experiments such as this. [3]

[Total: 15]

For Examiner's Use

2. Huntington's disease (HD) is a lethal neurodegenerative disease that is inherited as an autosomal dominant condition. As the appearance of symptoms may occur later in life (as late as at age 40), patients with HD may unknowingly pass on the mutant HD allele to their children.

A radioactive G8 probe revealed two RFLP variants that are linked to the HD allele which can be used in the screening of HD. Fig. 2.1 shows the two RFLP variants upstream to the HD allele and the G8 probe. The arrows show the position of *Hind*III restriction sites. The numbers represent kilobases.

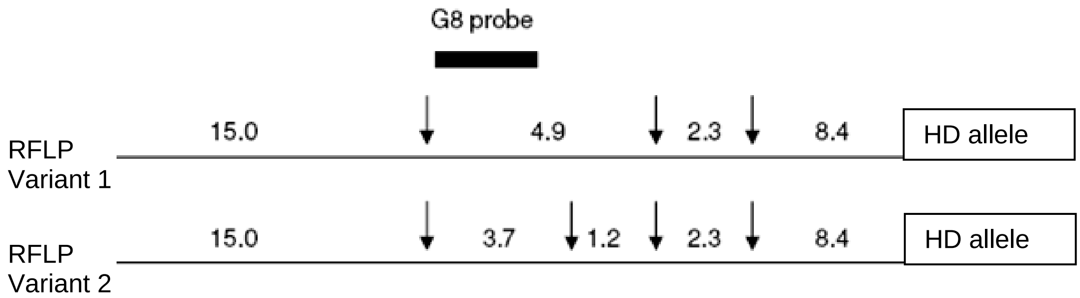

Fig. 2.1

Fig. 2.2 illustrates the Southern Blot analysis of the two RFLP variants in 30 related individuals with a family history of HD.

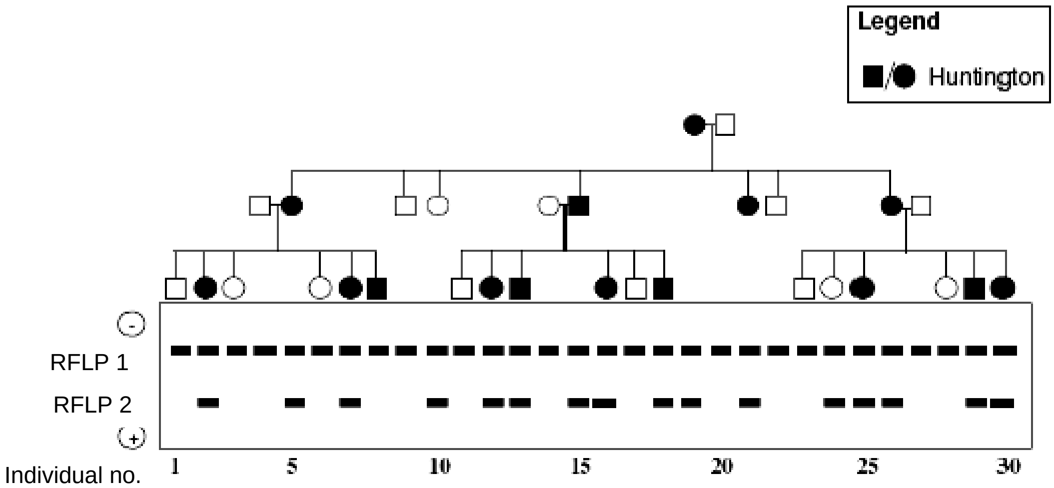

Fig. 2.2

For Examiner's Use

With reference to Fig. 2.1 and 2.2,

(a) outline how the DNA band patterns of the 30 individuals were obtained. [6]

(b) identify the RFLP variant which is linked to the mutant HD allele. Explain your answer. [2]

(c) suggest two possible reasons for the difference in the DNA band patterns among unaffected individuals. [4]

1.

2.

(d) discuss the effectiveness of using this RFLP marker in the screening of HD. [2]

[Total: 14]

For Examiner's Use

3.

(a) Outline the important features and the normal role of embryonic stem cells at the 4-cell stage in living organisms. [3]

(b) Heart attack causes damage to heart muscles. To repair damaged heart tissues, bone marrow stem cells are harvested from the patient, cultured and injected directly into the heart during coronary by-pass operation.

(i) Describe an important feature of the bone marrow stem cells that enables this to happen. [2]

(ii) Suggest how the bone marrow stem cells are coaxed to become healthy heart muscle tissue. [2]

(iii) Suggest why the bone marrow stem cells from the patient but not from other donors were used to repair the damaged heart muscles. [1]

For Examiner's Use

(c) About twenty percent of the DNA produced by PCR is replicated inaccurately.

Suggest and explain why it is not effective to use CFTR gene cloned by PCR in treating cystic fibrosis. [3]

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[Total: 11]

Planning Question

Write your answer to this question on the separate answer paper provided.

4. DNA is the genetic material in all living organisms. DNA from an external source can be taken up by bacteria by a process known as transformation.

Plan an experiment to investigate the effect of varying plasmid concentration on the efficiency of *E. coli* transformation.

You may, or otherwise, also use the following equipment and materials:

- Solution of plasmid containing ampicillin resistance gene of concentration 100 Unit/ml
- Suspension of competent *E. coli* cells
- Distilled water
- LB broth (nutrient solution)
- Petri dishes containing LB agar medium and ampicillin (LB/amp plates)
- Sterilised microfuge tubes
- Micropipettes
- Bleach solution
- alcohol

Your plan should have a clear and helpful structure to include:

- an explanation of theory to support your practical procedure
- a description of the method used including scientific reasoning behind the method
- an explanation of the dependent and independent variables involved
- relevant, clearly labeled diagrams
- how you will record your results and ensure they are as accurate and reliable as possible
- proposed layout of results tables and graphs with clear headings and labels
- correct use of scientific and technical terms

[Total: 12]

For Examiner's Use

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a), (b)** etc., as indicated in the question.

5. **(a)** The most common form of severe combined immunodeficiency disease (SCID) is caused by an X-linked mutation.

(i) Explain how X-linked SCID may be treated using a viral delivery system. [8]

(ii) Discuss the advantages and limitations of treatment using viral delivery method. [5]

(b) Discuss the benefits of the Human Genome Project (HGP) to human health. [7]

[Total: 20]

༄ End of Paper ༄